

# HPS Gaussian-Process Resonance Search

## Combined Global Search and Observed Limits

Analysis note v4.9.16: deficit extension

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6 September 2026

**Scope of the result.** Full 2015 and 2016 data, plus native 2021 10%, are combined over 19–250 MeV. The upper limits are pointwise, asymptotic 90% CL<sub>s</sub> limits. The GP calculation estimates scan-wide probabilities under one archived joint stress background. It does not establish a final discovery significance or calibrate confidence-interval coverage.

## 1 What the combined calculation adds

At each mass, the datasets that cover that mass share one possible particle coupling. Their information is combined in a single likelihood. Membership follows the established ranges; the analysis does not choose whichever dataset gives the strongest-looking result.

The upper curve limits the allowed particle contribution at each mass. The probability curves below it ask how often the chosen background would produce something this unusual somewhere in the entire search. The look-elsewhere calculation changes those probabilities and does not automatically rescale the upper limits. The principal profiled scan has no exceedance in 1,000 direct joint experiments. Its conditional probability is unresolved; the one-sided 95% upper bound is 0.003. The Gaussian field also has no sampled exceedance. An extreme stress-centered score is not an established particle significance.

| Statistic | Mass [MeV] | Local $p$              | GP global $p$          | Direct count |
|-----------|------------|------------------------|------------------------|--------------|
| Profiled  | 76         | $6.98 \times 10^{-20}$ | $<1.50 \times 10^{-5}$ | 0/1000       |
| Fixed     | 76         | $3.67 \times 10^{-13}$ | $<1.50 \times 10^{-5}$ | 0/1000       |

Table 1: Principal minimum-local-p ordering under the common stress background. Less-than signs are one-sided 95% Monte Carlo upper bounds, not measured zero probabilities. Fixed-background results are a separate diagnostic method.

| Mass [MeV] | Active data                      |
|------------|----------------------------------|
| 19–38      | 2015 full                        |
| 39–49      | 2015 full + 2016 full            |
| 50–90      | 2015 full + 2016 full + 2021 10% |
| 91–180     | 2016 full + 2021 10%             |
| 181–250    | 2021 10%                         |

Table 2: The complete search uses 1 MeV spacing and established dataset ranges.

## 2 Full combined search

### Combined HPS search over the full mass region

2015 full + 2016 full + 2021 10% | 19-250 MeV | 1 MeV grid

Global probabilities condition on one archived joint stress background.

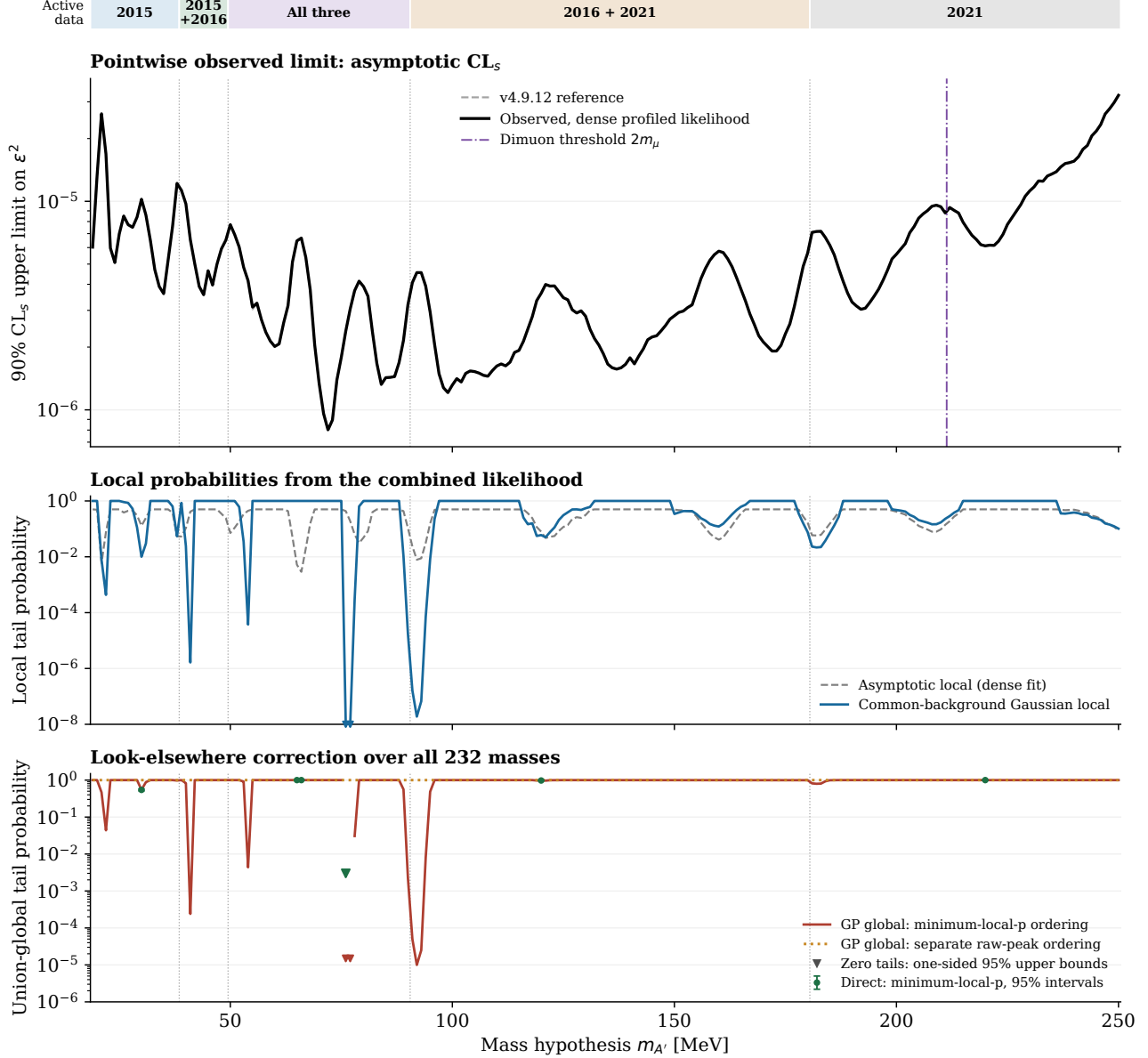


Figure 1: The complete 232-point search: the profiled observed upper limit above local and global probabilities. The activity strip fixes which data enter each mass. The upper curve includes the visible branching correction once above the dimuon threshold. Lower panels condition on one common joint stress background. Gray local probabilities use the dense observed root and the asymptotic convention; the Gaussian-response local rule assigns one to nonpositive raw roots. Local triangles mark values below the  $10^{-8}$  display floor. Global triangles are one-sided 95% Monte Carlo upper bounds for zero simulated tails, separated from probability estimates. Green points check the principal ordering; their error bars are central 95% intervals. The two global orderings are different tests. Expected-limit bands are not shown.

### 3 Illustrative scan of deficits

A negative auxiliary signal amplitude describes missing events with the signal-template shape; it is not a physical negative rate or coupling. This figure mirrors the excess scan without changing its fits or limits.

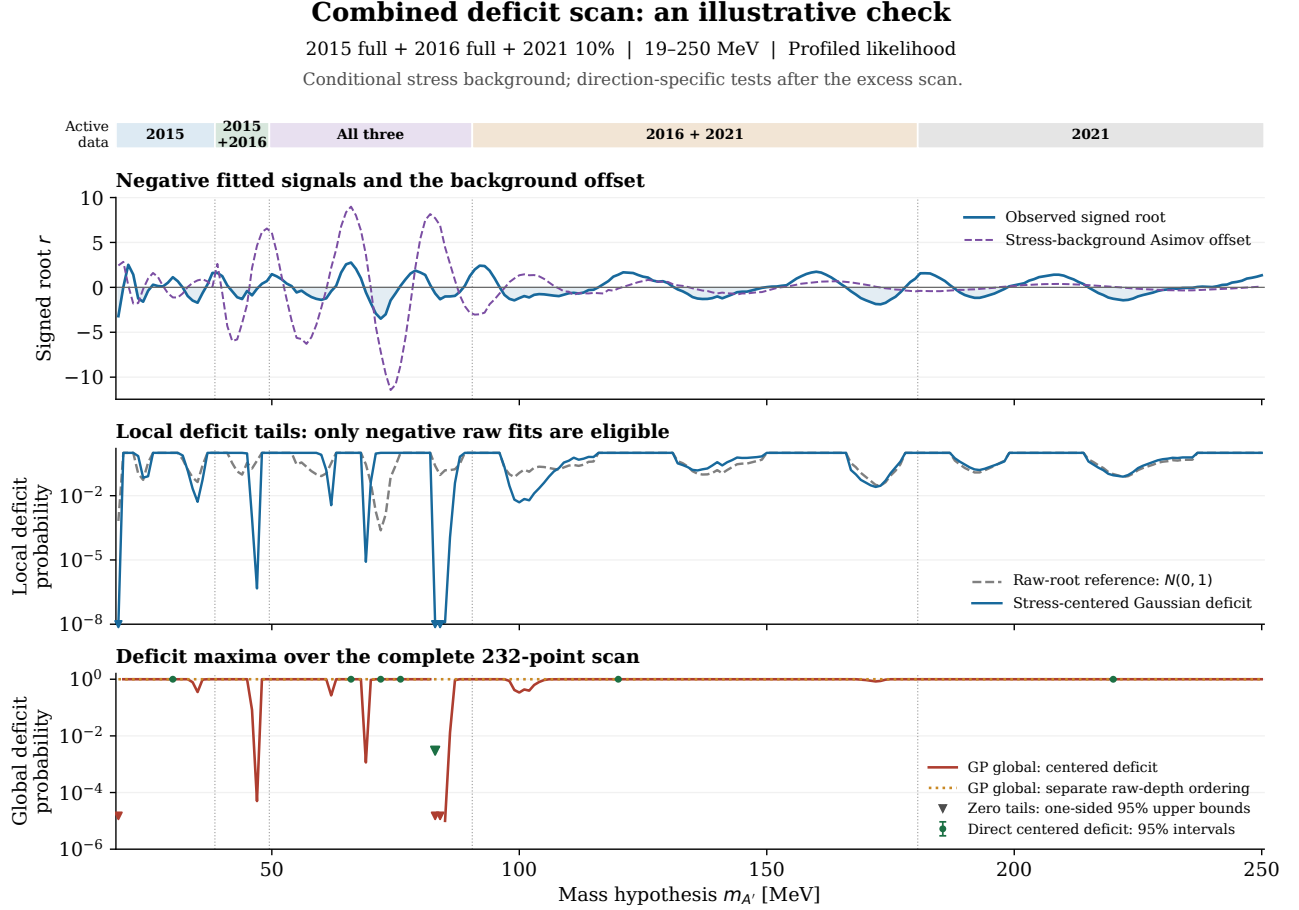


Figure 2: Profiled deficit scan. Top: observed signed roots and the stress-background offset; shading marks negative fitted contributions. Middle: raw-root Gaussian reference and conditional Gaussian-response tails. Both assign one to nonnegative raw roots. Local triangles mark the  $10^{-8}$  display floor. Bottom: two separate global orderings; triangles mark one-sided 95% zero-count bounds. Direct error bars are central 95% intervals.

For  $z = (r - a)/s$ , the conditional local rule is  $p^- = \Phi(z)$  when  $r < 0$ , and one otherwise. The principal scan score is  $T^- = \max_{m:r_m < 0}(-z_m)$ , with  $-\infty$  for an empty set. The separate raw-depth statistic is  $D^- = \max_m \max(0, -r_m)$ .

The deepest raw profiled deficit is at 72 MeV, with  $r = -3.490$  and an uncalibrated local reference of  $2.41 \times 10^{-4}$ . The largest stress-centered deficit instead occurs at 83 MeV:  $r = -0.676$ ,  $a = +7.707$  and  $s = 0.983$ . Its raw reference is 0.2495, whereas the conditional local value is  $7.18 \times 10^{-18}$ . This contrast again comes mainly from the stress offset.

The latter threshold has 0/200,000 GP and 0/1,000 direct exceedances. The one-sided 95% global upper bounds are  $1.50 \times 10^{-5}$  and 0.003, respectively. Every simulated raw-depth maximum exceeds the observed deficit. These are the same realizations used for the excess scan; no new fits or independent toys were needed.

These illustrative tails follow the excess scan and do not adjust for choosing directions, methods or orderings. The unresolved tails establish neither a particle signal nor physical background validity.

## 4 One shared-coupling likelihood

Let  $\mathcal{D}(m)$  be the predeclared active dataset set. The model combines Poisson bins and independent Gaussian background constraints:

$$\mathcal{L}_m(\epsilon^2, \theta) = \prod_{d \in \mathcal{D}(m)} \prod_{i \in W_d(m)} \text{Pois}(n_{di} \mid b_{di} + (L_d \theta_d)_i + \epsilon^2 S_{di}(m)) \prod_{d \in \mathcal{D}(m)} e^{-\theta_d^\top \theta_d / 2}.$$

$S_{di} = \text{BR}_{ee} T_{di}$  is the visible yield per unit physical coupling, where  $T_{di}$  is the stored unit-electron-branching template. The fitted coordinate is  $\epsilon_{\text{eff}}^2 = \epsilon_{\text{phys}}^2 \text{BR}_{ee}$ ; the display restores  $\epsilon_{\text{phys}}^2 = \epsilon_{\text{eff}}^2 / \text{BR}_{ee}$ . The implementation rescales this coordinate into a total window amplitude. No independent signal strength is fitted in each year.

An auxiliary unrestricted amplitude supplies the signed likelihood-ratio root,

$$r_m = \text{sign}(\hat{A}_m) \sqrt{2 \left[ \text{NLL}_m(A=0) - \text{NLL}_m(\hat{A}_m) \right]}.$$

Here  $\text{NLL}_m(A)$  is the negative log likelihood minimized over the background nuisances at fixed  $A$ . Poisson expectations remain positive. Physical discovery and upper-limit tests retain the nonnegative signal boundary; the auxiliary negative branch is needed to model the signed field.

### Complete joint experiments

Ten pilot and 1,000 separate validation spectra per year are reused from v4.9.14/v4.9.15. The year-specific random streams are distinct. Pairing equal row indices gives 1,000 joint experiments, not 3,000. Each experiment supplies exactly the same full spectrum at every mass in which its dataset participates.

Kernels, templates, support and exclusion masks remain frozen. Every toy retrains its log-count background prediction and count-dependent training errors. All multi-dataset intervals receive new common-amplitude fits. Only the single-dataset intervals reuse existing likelihood-root columns.

The generating means are the archived smooth stress spectra. Event-level independence of the 2016 development input from its full observed sample has not been established. Independence here describes the conditional toy model; omitted experimental systematic correlations have not been newly quantified.

### One response basis across boundaries

Following Ananiev and Read [1], perturb each data bin by its Poisson standard deviation. This extension retains the nonzero offset and response width:

$$a_m = r_m(B), \quad D_{im} = r_m(B + \sqrt{B_i} e_i) - a_m, \quad C = D^\top D, \quad s_m = \sqrt{C_{mm}}, \quad K_{mn} = \frac{C_{mn}}{s_m s_n}.$$

The basis has 484, 720 and 422 bins from 2015, 2016 and 2021. An inactive dataset has exactly zero response at that mass. This retains correlations across boundaries that share data; separately sampled segment maxima are not joined.

For each statistic, 200,000 vectors  $Z \sim N(0, K)$  generate  $r_m^* = a_m + s_m Z_m$ . The principal maximum is  $T = \max_{m: r_m^* > 0} Z_m$  (or  $-\infty$  for an empty set); the secondary maximum is  $R = \max_m \max(0, r_m^*)$ . Their respective tails are compared with the independent joint Poisson scans.

## 5 Offsets and widths of the combined statistic

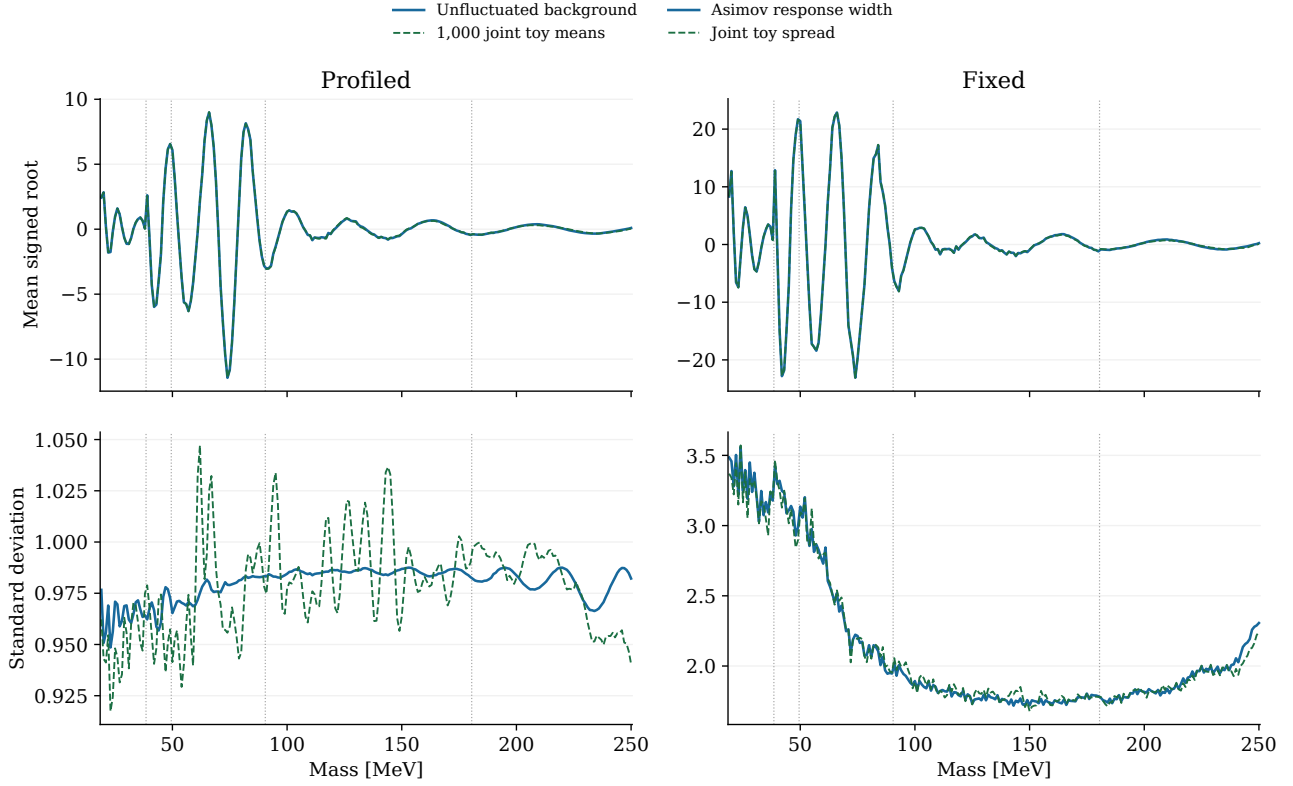


Figure 3: Asimov-response predictions compared with 1,000 joint Poisson scans. Top panels show the offset in the signed root; bottom panels show its spread. Vertical lines mark membership changes. Agreement tests the approximation under the generating means, without establishing their physical validity.

| Statistic | Standardized mean range | Spread range   | Flags |
|-----------|-------------------------|----------------|-------|
| Profiled  | -0.079 to +0.076        | 0.957 to 1.074 | 0     |
| Fixed     | -0.093 to +0.080        | 0.943 to 1.050 | 0     |

Table 3: Normality flags use a Holm adjustment across 232 masses within each method. Non-rejection has finite power and cannot certify extreme Gaussian tails.

Centering, scaling and covariance come from deterministic bin perturbations. Validation scans do not tune them. Profiled and fixed-background statistics use identical spectra but can have different sampling widths.

Large oscillating offsets can make an ordinary observed value unusual relative to a particular stress construction. A small probability can diagnose that construction without identifying a particle. The inherited 2016 numerical exception, source-fit waiver, development overlap and 75–85 MeV source transition remain unresolved by this extension.

## 6 Correlations and the look-elsewhere check

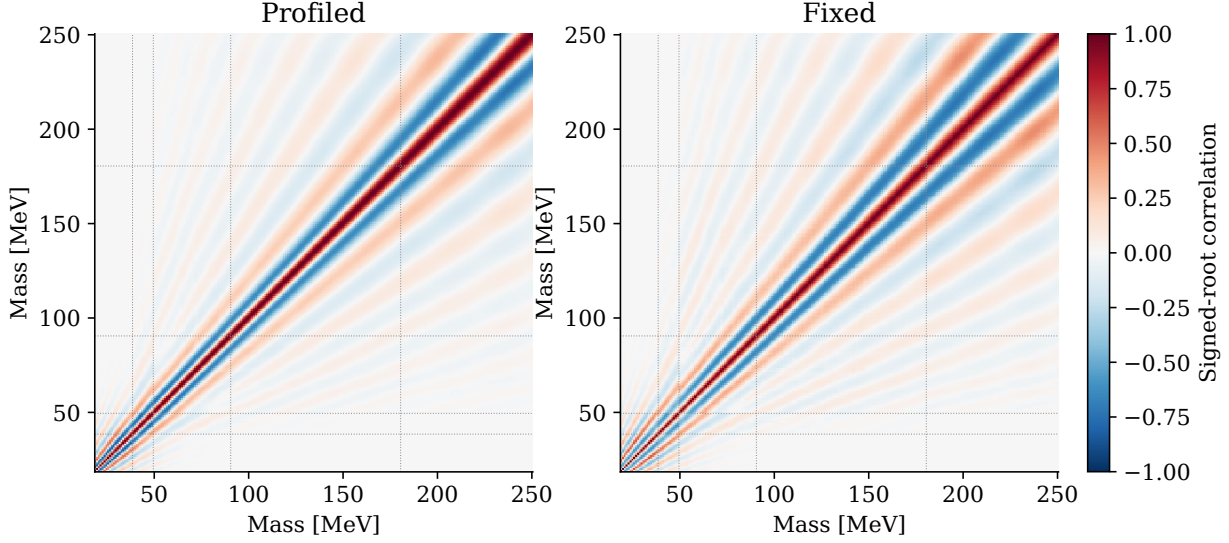


Figure 4: Response correlations across the union grid. Dotted lines show membership changes. Shared datasets retain correlations across boundaries; regions with no common dataset have zero response covariance in this model.

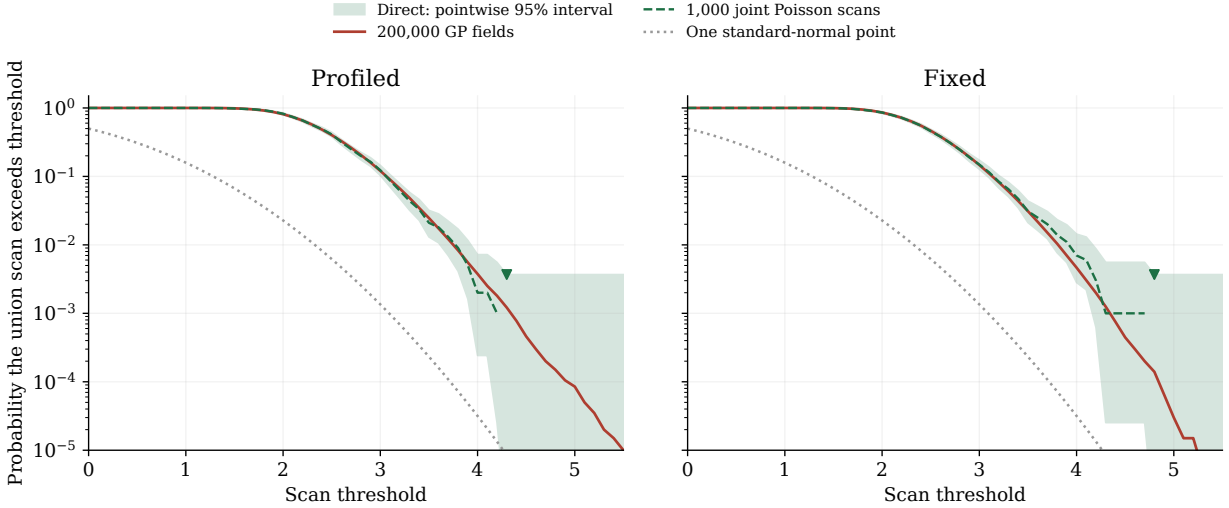


Figure 5: Principal scan-maximum tails. Direct-toy bands are pointwise central 95% binomial intervals, not expected-limit bands or simultaneous coverage bands. Triangles mark the upper endpoint of the central interval at the first unresolved direct tail. More GP draws reduce sampling error within the approximation, without replacing direct validation.

Profiled: the maximum-distribution KS diagnostic gives nominal  $p=0.8625$  for the principal ordering and 0.6389 for the raw ordering. The correlation RMS difference is 0.031.

Fixed: the maximum-distribution KS diagnostic gives nominal  $p=0.9974$  for the principal ordering and 0.2034 for the raw ordering. The correlation RMS difference is 0.030.

These diagnostics compare simulation methods, not the data significance. Two methods are checked per ordering. Direct Poisson counts calibrate the declared score under the generating spectrum without requiring a jointly Gaussian field; the GP estimate has that additional approximation.

## 7 Representative probabilities

For positive raw roots, the principal local rule is  $p_m = \bar{\Phi}[(r_m - a_m)/s_m]$ ; otherwise  $p_m = 1$ . Its global curve counts how often any of the 232 masses reaches that local threshold. The separate raw ordering scans  $\max_m \max(0, r_m)$ . These tests cannot be selected according to which gives the smaller probability.

| Mass | Active years   | Asymp. local | GP local               | GP global              | Direct    | Raw global |
|------|----------------|--------------|------------------------|------------------------|-----------|------------|
| 30   | 2015           | 0.1291       | 0.01                   | 0.5479                 | 543/1000  | 1          |
| 65   | 2015+2016+2021 | 0.0052       | 1                      | 1                      | 1000/1000 | 1          |
| 66   | 2015+2016+2021 | 0.0029       | 1                      | 1                      | 1000/1000 | 1          |
| 76   | 2015+2016+2021 | 0.4342       | $6.98 \times 10^{-20}$ | $<1.50 \times 10^{-5}$ | 0/1000    | 1          |
| 120  | 2016+2021      | 0.0795       | 0.0598                 | 0.9813                 | 975/1000  | 1          |
| 220  | 2021           | 0.5          | 1                      | 1                      | 1000/1000 | 1          |

Table 4: Profiled representative masses: the principal and raw extrema, plus 30, 65, 120 and 220 MeV. GP local uses the common-background Gaussian response. Both global columns use the full union at the threshold observed at that mass. Raw global is the separate raw-peak test.

| Statistic | Mass | Raw $r$ | Offset $a$ | Width $s$ | $(r - a)/s$ |
|-----------|------|---------|------------|-----------|-------------|
| Profiled  | 76   | +0.166  | -8.700     | 0.979     | +9.053      |
| Fixed     | 76   | +0.175  | -15.348    | 2.164     | +7.173      |

Table 5: Principal peak decomposition. The final column is a conditional score, not an independently established discovery significance.

At the profiled principal peak, the unadjusted asymptotic local probability is 0.4342, while centering and scaling against the stress background gives  $6.98 \times 10^{-20}$ . The observed raw root is close to zero; the extreme centered score arises mainly because the stress background predicts a large negative root.

At 74 and 75 MeV, the centered scores are 10.20 and 10.38, but the raw roots are negative. The declared boundary therefore assigns local  $p=1$  there. The next positive raw root becomes the principal peak. This abrupt change comes from the signal boundary together with the stress offset; it is not a numerical sign flip.

The raw maximum occurs at 66 MeV. All 200,000 GP fields and all 1,000 direct experiments exceed it. These differences reflect the specified orderings and background offsets; they are not a free choice of significance. The raw maximum tests positive peaks; it does not assess full-spectrum goodness of fit.

| Statistic | Full 1 MeV grid        | First 2 MeV subgrid    | Second subgrid         |
|-----------|------------------------|------------------------|------------------------|
| Profiled  | $<1.50 \times 10^{-5}$ | $<1.50 \times 10^{-5}$ | $<1.50 \times 10^{-5}$ |
| Fixed     | $<1.50 \times 10^{-5}$ | $<1.50 \times 10^{-5}$ | $<1.50 \times 10^{-5}$ |

Table 6: GP probabilities at the same original peak threshold. Zero-count bounds do not establish fine-tail convergence.

Subgrids remove alternate masses at the original observed threshold. They test finite-grid dependence, not convergence to a continuous search. Probabilities include neither a choice between methods nor uncounted searches over widths, kernels or other analysis choices.

## 8 Observed limits and the v4.9.12 comparison

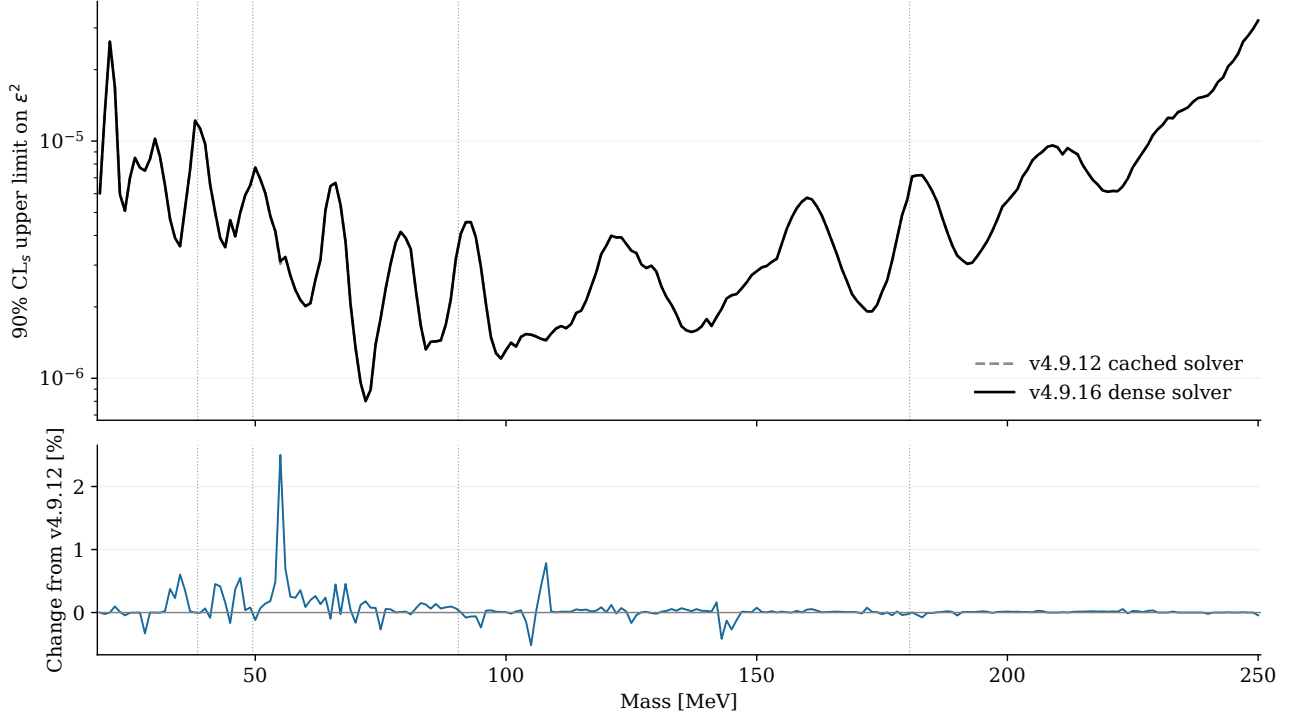


Figure 6: The pointwise upper-limit construction with the current dense and earlier cached solvers. The lower panel shows their relative difference. Both curves have the same membership and branching correction. Their difference is not a look-elsewhere adjustment or a sensitivity measurement.

The largest absolute relative change from the corresponding v4.9.12 limit is 2.502%. The largest difference in the nonnegative discovery root is 0.0267. Fresh dense fits reproduce v4.9.13 dense references where the same scope exists. The old optimizer does not select a preferred answer. Above 211.317 MeV, displayed limits include  $1 + \sqrt{1 - 4m_\mu^2/m^2}(1 + 2m_\mu^2/m^2)$  once; the CSV retains the electron-channel values.

The upper limit inverts the bounded, piecewise-asymptotic  $CL_s$  statistic at each mass [2]. The background-only joint scans test the discovery-score approximation. They do not supply the signal-plus-background distributions needed to calibrate the full  $CL_s$  curve. A GP global probability is not a replacement  $CL_s$  tail.

The v4.9.13 study contains conditional calibrated individual limits and the common all-three interval. It has no calibrated pairwise endpoints over the remaining union. Splicing different calibrations would conceal a change of construction. A fully calibrated union limit requires additional declared signal-plus-background work; this note supplies the complete observed asymptotic curve.

## 9 Numerical checks, provenance and continuation

All 2736 numerical and product-identity checks passed. There are 142 newly fitted joint coordinates and 90 reused single-dataset coordinates. Summed coordinate execution took 24.1 minutes with one worker and one BLAS thread. The exact backend was retained at 1 new coordinates (41 MeV). The largest paired pilot-root difference is  $2.10 \times 10^{-5}$ , the largest complete-response relative vector error is  $4.73 \times 10^{-6}$ , and the largest sentinel-correlation error is  $4.65 \times 10^{-7}$ . No paired pilot changed its raw-positive classification.

Every new joint mass has ten paired exact pilot fits, an exact baseline and a fixed bin-response stencil. Complete exact responses are retained at 39, 49, 50, 90, 91 and 180 MeV. The accelerator must pass root, centered-response, width and correlation gates. Otherwise all ensembles at that coordinate use the exact backend. Failed exact references halt; no toy or hypothesis is removed.

The v4.9.14 and v4.9.15 inputs are checked against their full frozen manifests. The v4.9.12 comparison table is checked against its specific frozen-manifest entry. The isolated derivative is

```
study_results/v4p9p16_combined_global_20260906/
```

It retains spectra, joint root vectors, limits, covariance, GP maxima, probability tables and per-mass audits. The independent HEP review is under `review/`. The original manifest remains unchanged. The deficit extension and this revised report have a separate manifest under

```
study_results/v4p9p16_deficit_extension_20260906/
```

The independent deficit audit passed 939 conditions.

From the repository root, verify the completed numerical products with

```
python3 -B \  
study_results/v4p9p16_combined_global_20260906/verify_combined.py
```

Preserve the finished version before reanalysis or new generation. `NEXT_STEPS.md` gives resume commands, new-seed requirements and the additional work for calibrated limits.

The next scientific priority is to qualify the joint background family with predeclared predictive controls. Direct tail precision, finer grids and signal injections answer separate questions. Additional GP sampling alone cannot establish physical background validity, expected sensitivity or interval coverage.

## References

- [1] V. Ananiev and A. L. Read, *Gaussian Process-based calculation of look-elsewhere trials factor*, JINST **18** (2023) P05041. arXiv:2206.12328v3.
- [2] G. Cowan, K. Cranmer, E. Gross and O. Vitells, *Asymptotic formulae for likelihood-based tests of new physics*, Eur. Phys. J. C **71** (2011) 1554. arXiv:1007.1727.